

5A Suboi Zenabu first version.docx

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ASSIGNMENT 5A – METHODS

1. Title

- a. Drivers of ³Impact of Malaria Interventions in Ghana: A Global Sensitivity Analysis
- b. Understanding the Key Drivers of ³Impact of Malaria Interventions in Ghana
- c. A Sensitivity Analysis to Determine the Key Determinants of Malaria Intervention Impact at the District Level in Ghana

2. Methods

Key:

Overview, research aim, and location

Subjects and data

procedure

This study used mathematical modeling of malaria transmission, Gaussian process emulators, and a variance-based sensitivity approach to identify the key drivers of impact of malaria interventions on ²malaria incidence at the district level in Ghana.

To simulate ²malaria incidence at the district level in Ghana, we used OpenMalaria v44.0 (1), an individual-based model (IBM) of ¹malaria transmission. This stochastic open-source model was developed by the **disease modeling unit of the Swiss Tropical and Public Health Institute**. It comprises an individual-based human model and a mechanistic mosquito dynamics model for simulating malaria epidemiology and transmission in various transmission settings. To obtain malaria incidence in Ghana that reflects district-specific intervention combinations as specified in the **National Malaria Elimination Strategic Plan (NMESP) (2)**, we parameterized the OpenMalaria model using district-level historical intervention coverage and epidemiological data. We used a 10,000 human population size with age structure from health and demographic surveillance data of Ghana. Since OpenMalaria is a complex model with many parameters, it is computationally intensive to simulate malaria transmission; hence, we implemented the Latin hypercube sampling (LHS) method to reduce the number of model runs.

Having obtained key metrics of interest (ie, malaria incidence) from the OpenMalaria model, we trained a Gaussian process (GP) surrogate model for each of the 261 districts in Ghana. The GP surrogate model learned the input-output relationship between the input and output of the OpenMalaria and was used as a faster alternative during the sensitivity analysis. To quantify the contribution of each model input parameter to the output variability, we used the

GP model and a variance-based sensitivity analysis method, such as the Sobol method (3), to systematically evaluate the relative importance of key intervention-related parameters on malaria incidence reduction from 2023 to 2026. Since such methods require rapid and reliable prediction of model outcomes across a high-dimensional parameter space, introducing the GP model in this method allowed us to examine how variations in parameters, such as intervention coverage, contribute to reductions in malaria incidence at the district level.

3. Acknowledgments

a. Research paper

We thank all our collaborators for their expertise and time dedicated to this project. Special thanks to all the Analytics and Interventions Modeling (AIM) group members of the Swiss Tropical and Public Health Institute for the very insightful discussion sessions we had with them. We thank Ghana's National Malaria Elimination Programme (NMEP) for granting us access to district-level demographic, historical coverage of interventions, and epidemiological data.

b. Thesis

I want to thank my supervisors for their guidance and patience toward me during this project. I also want to express my sincere gratitude to Dr. Emilie Pothin for releasing funds to support this study and all members of the Analytics and Interventions Modeling (AIM) group of the Swiss Tropical and Public Health Institute for providing a supportive environment for this project.

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